AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	13.9 / 2.23517 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-08-01 06:00 UTC
Location	29.6957, -95.4992
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 858	34.21 to 100; mean 71.3818	83.7 at 2026-08-01 05:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 858	23.3333 to 36.3889; mean 30.0437	27 at 2026-08-01 05:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1166	0 to 360; mean 190.652	220 at 2026-08-01 05:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1205	0 to 8.2311; mean 3.0009	3.60111 at 2026-08-01 05:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 130	5879.03 to 5955.28; mean 5926.7	5945.53 at 2026-08-01 06:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 130	58.0634 to 2520.22; mean 917.524	540.541 at 2026-08-01 06:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 130	28.4162 to 57.7464; mean 42.7388	39.0798 at 2026-08-01 06:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 130	1001.37 to 1014.57; mean 1009.5	1010.73 at 2026-08-01 06:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 130	18.6534 to 22.6586; mean 20.8164	21.9846 at 2026-08-01 06:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 130	-3.3353 to 4.5116; mean 1.3504	1.61858 at 2026-08-01 06:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 130	-3.70918 to 6.59256; mean 1.5743	3.73033 at 2026-08-01 06:00Z; at event time
openaq	Ozone (ozone)	ppm	17 / 1120	-0.003 to 0.075; mean 0.0203	0.013 at 2026-08-01 06:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 214	11 to 121; mean 29.1776	13 at 2026-08-01 06:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	70 / 3014	0 to 30.9; mean 5.948	0.3 at 2026-08-01 06:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 361	0 to 100; mean 52.3989	1 at 2026-08-01 06:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 361	43 to 96; mean 72.6898	83 at 2026-08-01 06:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 361	0 to 0.22; mean 0.0009	0 at 2026-08-01 06:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 361	1007 to 1016; mean 1012.6	1014 at 2026-08-01 06:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 361	25.25 to 37.63; mean 30.9571	27.26 at 2026-08-01 06:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 361	0 to 351; mean 188.355	179 at 2026-08-01 06:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 361	0.45 to 6.22; mean 2.296	0.45 at 2026-08-01 06:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	72 / 5154	0 to 313; mean 8.0963	3.7 at 2026-08-01 06:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0274288 to 0.0286852; mean 0.0279	0.0286852 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	4 / 4	0.000197921 to 0.000219694; mean 0.0002	0.000219694 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	3.93606e-05 to 4.98019e-05; mean 0	4.98019e-05 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	4 / 4	6.51867e-06 to 3.45966e-05; mean 0	7.83617e-06 at 2026-07-31 19:45Z; 10 h 15 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-31 19:45Z; 10 h 15 min before event
tceq	Carbon monoxide (co)	ppm	3 / 188	0 to 0.7; mean 0.1846	0.2 at 2026-08-01 06:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 792	-2.7 to 22.7; mean 3.6244	13.9 at 2026-08-01 06:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 187	-0.7 to 4.4; mean -0.0225	-0.6 at 2026-08-01 06:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-30 18Z 50.99, 07-31 00Z 77.28, 06Z 90.36, 12Z 70.88, 18Z 47.73, 08-01 00Z 76.11, 06Z 92.21, 12Z 75.08, 18Z 50.92, 08-02 00Z 76.09, 06Z 87.78, 12Z 61.19
asos	Air temperature (temperature)	07-30 18Z 34.32, 07-31 00Z 28.59, 06Z 25.59, 12Z 30.09, 18Z 34.6, 08-01 00Z 28.63, 06Z 25.83, 12Z 30.08, 18Z 34, 08-02 00Z 29.89, 06Z 27.78, 12Z 31.07
asos	Wind direction (wind_direction)	07-30 18Z 182.7, 07-31 00Z 171.8, 06Z 141.6, 12Z 209.9, 18Z 184.5, 08-01 00Z 192.1, 06Z 176.3, 12Z 237.1, 18Z 237.8, 08-02 00Z 204.1, 06Z 233.7, 12Z 121.5
asos	Wind speed (wind_speed)	07-30 18Z 3.84, 07-31 00Z 3.349, 06Z 1.584, 12Z 3.208, 18Z 3.205, 08-01 00Z 3.835, 06Z 2.24, 12Z 4.049, 18Z 3.282, 08-02 00Z 2.812, 06Z 2.646, 12Z 1.957
noaa_gfs	500-hPa geopotential height (gh_500)	07-30 18Z 5941, 07-31 00Z 5940, 06Z 5950, 12Z 5941, 18Z 5954, 08-01 00Z 5948, 06Z 5946, 12Z 5927, 18Z 5927, 08-02 00Z 5905, 06Z 5902, 12Z 5881, 18Z 5887
noaa_gfs	Planetary boundary layer height (pbl_height)	07-30 18Z 1674, 07-31 00Z 879.6, 06Z 382.9, 12Z 266, 18Z 1679, 08-01 00Z 985, 06Z 413.4, 12Z 370.9, 18Z 1660, 08-02 00Z 942, 06Z 304.5, 12Z 186.4, 18Z 2185
noaa_gfs	Precipitable water (precipitable_water)	07-30 18Z 41.51, 07-31 00Z 44.5, 06Z 36.3, 12Z 39, 18Z 41.28, 08-01 00Z 43, 06Z 38.67, 12Z 42.01, 18Z 50.44, 08-02 00Z 56.41, 06Z 51, 12Z 39.75, 18Z 31.74
noaa_gfs	Surface pressure (surface_pressure)	07-30 18Z 1011, 07-31 00Z 1009, 06Z 1012, 12Z 1012, 18Z 1012, 08-01 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1009, 08-02 00Z 1005, 06Z 1007, 12Z 1008, 18Z 1008
noaa_gfs	850-hPa temperature (t_850)	07-30 18Z 19.29, 07-31 00Z 20.97, 06Z 22.12, 12Z 21.59, 18Z 19.24, 08-01 00Z 21.16, 06Z 22.21, 12Z 21.96, 18Z 19.7, 08-02 00Z 21.62, 06Z 20.8, 12Z 20.45, 18Z 19.5
noaa_gfs	10-m eastward wind (u_10m)	07-30 18Z 0.9009, 07-31 00Z -0.6853, 06Z 0.7154, 12Z 1.314, 18Z 1.646, 08-01 00Z 0.07614, 06Z 1.479, 12Z 2.217, 18Z 4.054, 08-02 00Z 1.988, 06Z 2.851, 12Z 1.131, 18Z -0.1314
noaa_gfs	10-m northward wind (v_10m)	07-30 18Z 2.503, 07-31 00Z 4.724, 06Z 3.255, 12Z 1.941, 18Z 2.424, 08-01 00Z 3.582, 06Z 3.25, 12Z 1.789, 18Z 0.5916, 08-02 00Z -0.2505, 06Z 1.536, 12Z -1.846, 18Z -3.033
openaq	Ozone (ozone)	07-30 18Z 0.03185, 07-31 00Z 0.0173, 06Z 0.01067, 12Z 0.01455, 18Z 0.03296, 08-01 00Z 0.01706, 06Z 0.01189, 12Z 0.01401, 18Z 0.03069, 08-02 00Z 0.02191, 06Z 0.009128, 12Z 0.02412, 18Z 0.05763
openaq	PM10 (pm10)	07-30 18Z 37.73, 07-31 00Z 26.78, 06Z 22.39, 12Z 48.72, 18Z 37.61, 08-01 00Z 32.12, 06Z 16.94, 12Z 30.22, 18Z 31.83, 08-02 00Z 24.33, 06Z 22.22, 12Z 22.89, 18Z 17.67
openaq	PM2.5 (pm25)	07-30 18Z 5.051, 07-31 00Z 5.375, 06Z 6.231, 12Z 6.995, 18Z 5.701, 08-01 00Z 3.892, 06Z 3.225, 12Z 4.784, 18Z 5.638, 08-02 00Z 8.724, 06Z 10.03, 12Z 8.118, 18Z 6.561
openweather	Cloud cover (cloud_cover)	07-30 18Z 79.32, 07-31 00Z 85.33, 06Z 89.3, 12Z 86, 18Z 36.53, 08-01 00Z 5.633, 06Z 4.8, 12Z 13.1, 18Z 18.45, 08-02 00Z 55.87, 06Z 67.31, 12Z 85.45
openweather	Relative humidity (humidity)	07-30 18Z 56.52, 07-31 00Z 73.87, 06Z 87.7, 12Z 77.4, 18Z 52.7, 08-01 00Z 72.7, 06Z 87.5, 12Z 81.24, 18Z 55.77, 08-02 00Z 71.87, 06Z 85.97, 12Z 70.94
openweather	Precipitation rate (precipitation)	07-30 18Z 0, 07-31 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-01 00Z 0, 06Z 0, 12Z 0, 18Z 0.003871, 08-02 00Z 0, 06Z 0.007586, 12Z 0
openweather	Surface pressure (pressure)	07-30 18Z 1013, 07-31 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1014, 08-01 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1010, 08-02 00Z 1009, 06Z 1010, 12Z 1012
openweather	Air temperature (temperature)	07-30 18Z 35.77, 07-31 00Z 30.19, 06Z 26.54, 12Z 29.74, 18Z 36.1, 08-01 00Z 30.33, 06Z 26.73, 12Z 29.56, 18Z 35.64, 08-02 00Z 31.39, 06Z 28.4, 12Z 30.66
openweather	Wind direction (wind_direction)	07-30 18Z 188.2, 07-31 00Z 175.3, 06Z 158, 12Z 202, 18Z 182.6, 08-01 00Z 190, 06Z 154.6, 12Z 232.2, 18Z 264, 08-02 00Z 230.7, 06Z 194.6, 12Z 90.61
openweather	Wind speed (wind_speed)	07-30 18Z 2.143, 07-31 00Z 2.578, 06Z 2.035, 12Z 2.612, 18Z 2.057, 08-01 00Z 2.506, 06Z 2.446, 12Z 2.695, 18Z 2.235, 08-02 00Z 2.286, 06Z 2.043, 12Z 1.942

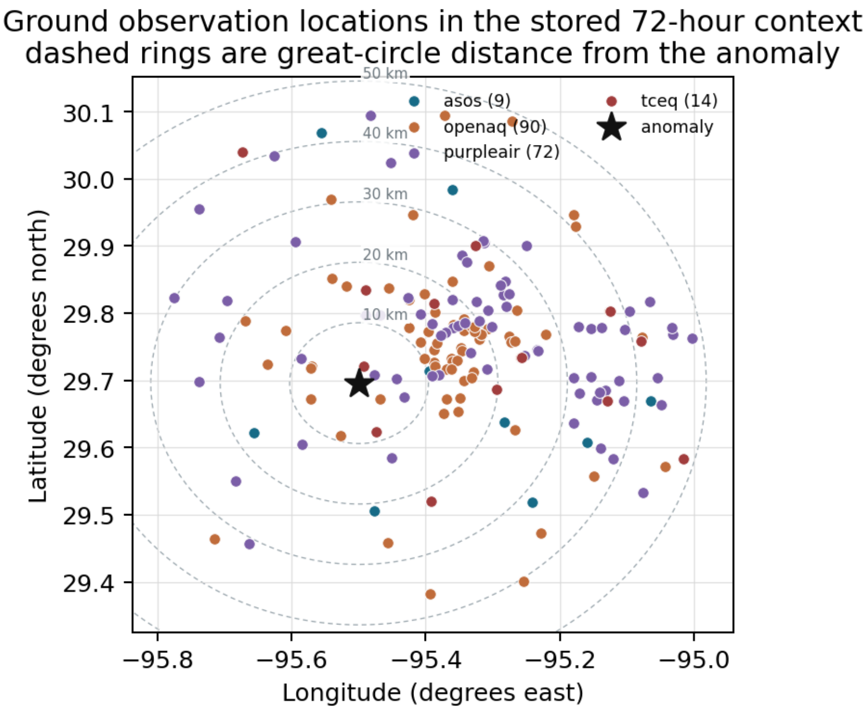
Source	Metric	Values
purpleair	PM2.5 (pm25)	07-30 18Z 6.493, 07-31 00Z 7.353, 06Z 8.37, 12Z 9.318, 18Z 6.62, 08-01 00Z 6.066, 06Z 4.321, 12Z 6.299, 18Z 7.23, 08-02 00Z 12.42, 06Z 12.23, 12Z 10.37, 18Z 12.5
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
sentinel5p	CO column (s5p_co_column)	07-30 18Z 0.02743, 07-31 18Z 0.02868, 08-01 18Z 0.02751
sentinel5p	CO granules available (s5p_co_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-30 18Z 0.0002134, 07-31 18Z 0.000219, 08-01 18Z 0.0001979
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-30 18Z 3.936e-05, 07-31 18Z 4.98e-05, 08-01 18Z 4.018e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-30 18Z 2.458e-05, 07-31 18Z 2.122e-05, 08-01 18Z 6.519e-06
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-30 18Z 1, 07-31 18Z 1, 08-01 18Z 1
tceq	Carbon monoxide (co)	07-30 18Z 0.1944, 07-31 00Z 0.1778, 06Z 0.2111, 12Z 0.2333, 18Z 0.2056, 08-01 00Z 0.1889, 06Z 0.1611, 12Z 0.2056, 18Z 0.1812, 08-02 00Z 0.22, 06Z 0.1167, 12Z 0.1333
tceq	Nitrogen dioxide (no2)	07-30 18Z 3.656, 07-31 00Z 2.06, 06Z 4.258, 12Z 4.728, 18Z 3.976, 08-01 00Z 2.005, 06Z 3.228, 12Z 3.299, 18Z 4.018, 08-02 00Z 5.772, 06Z 2.893, 12Z 3.138, 18Z 2.238
tceq	Sulfur dioxide (so2)	07-30 18Z -0.1556, 07-31 00Z -0.1444, 06Z -0.2, 12Z 0.3556, 18Z 0.4444, 08-01 00Z -0.08889, 06Z -0.1944, 12Z -0.1167, 18Z -0.04706, 08-02 00Z -0.18, 06Z -0.07778, 12Z -0.09444, 18Z 0.1

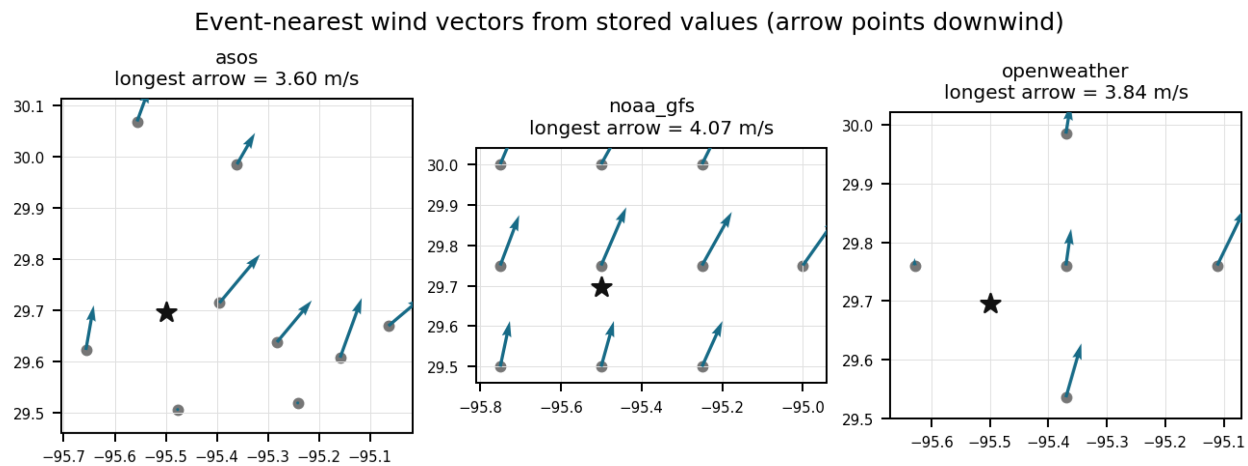
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (10.269 km) 69.78, SGR (17.253 km) 75.63, AXH (21.198 km) 71.52, HOU (21.928 km) 71.01, LVJ (31.726 km) 72.5, EFD (34.344 km) 71.72, IAH (34.767 km) 66.73, DWH (41.756 km) 72.58, +1 more
asos	Air temperature (temperature)	MCJ (10.269 km) 30.2, SGR (17.253 km) 29.74, AXH (21.198 km) 29.63, HOU (21.928 km) 30.27, LVJ (31.726 km) 29.9, EFD (34.344 km) 30.23, IAH (34.767 km) 30.94, DWH (41.756 km) 29.54, +1 more
asos	Wind direction (wind_direction)	MCJ (10.269 km) 211.3, SGR (17.253 km) 203.3, AXH (21.198 km) 130.1, HOU (21.928 km) 205.8, LVJ (31.726 km) 171.1, EFD (34.344 km) 187.1, IAH (34.767 km) 206.7, DWH (41.756 km) 194.7, +1 more
asos	Wind speed (wind_speed)	MCJ (10.269 km) 3.544, SGR (17.253 km) 3.28, AXH (21.198 km) 1.349, HOU (21.928 km) 3.776, LVJ (31.726 km) 2.251, EFD (34.344 km) 3.849, IAH (34.767 km) 3.144, DWH (41.756 km) 2.447, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.50 (6.033 km) 5927, gfs:29.50,-95.50 (21.766 km) 5927, gfs:29.75,-95.25 (24.811 km) 5926, gfs:29.75,-95.75 (24.957 km) 5927, gfs:29.50,-95.25 (32.471 km) 5926, gfs:29.50,-95.75 (32.583 km) 5927, gfs:30.00,-95.50 (33.831 km) 5927, gfs:30.00,-95.25 (41.501 km) 5926, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.50 (6.033 km) 1093, gfs:29.50,-95.50 (21.766 km) 857.8, gfs:29.75,-95.25 (24.811 km) 1047, gfs:29.75,-95.75 (24.957 km) 739.3, gfs:29.50,-95.25 (32.471 km) 767.4, gfs:29.50,-95.75 (32.583 km) 853.1, gfs:30.00,-95.50 (33.831 km) 1129, gfs:30.00,-95.25 (41.501 km) 1063, +2 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.50 (6.033 km) 42.97, gfs:29.50,-95.50 (21.766 km) 43.54, gfs:29.75,-95.25 (24.811 km) 43.77, gfs:29.75,-95.75 (24.957 km) 41.49, gfs:29.50,-95.25 (32.471 km) 44.06, gfs:29.50,-95.75 (32.583 km) 42.68, gfs:30.00,-95.50 (33.831 km) 41.7, gfs:30.00,-95.25 (41.501 km) 42.95, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.50 (6.033 km) 1010, gfs:29.50,-95.50 (21.766 km) 1010, gfs:29.75,-95.25 (24.811 km) 1011, gfs:29.75,-95.75 (24.957 km) 1008, gfs:29.50,-95.25 (32.471 km) 1011, gfs:29.50,-95.75 (32.583 km) 1010, gfs:30.00,-95.50 (33.831 km) 1008, gfs:30.00,-95.25 (41.501 km) 1009, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.50 (6.033 km) 20.78, gfs:29.50,-95.50 (21.766 km) 20.78, gfs:29.75,-95.25 (24.811 km) 20.67, gfs:29.75,-95.75 (24.957 km) 20.95, gfs:29.50,-95.25 (32.471 km) 20.75, gfs:29.50,-95.75 (32.583 km) 20.83, gfs:30.00,-95.50 (33.831 km) 20.94, gfs:30.00,-95.25 (41.501 km) 20.72, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.50 (6.033 km) 1.337, gfs:29.50,-95.50 (21.766 km) 1.061, gfs:29.75,-95.25 (24.811 km) 1.168, gfs:29.75,-95.75 (24.957 km) 1.729, gfs:29.50,-95.25 (32.471 km) 1.282, gfs:29.50,-95.75 (32.583 km) 1.622, gfs:30.00,-95.50 (33.831 km) 1.931, gfs:30.00,-95.25 (41.501 km) 0.8002, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.50 (6.033 km) 1.734, gfs:29.50,-95.50 (21.766 km) 2.033, gfs:29.75,-95.25 (24.811 km) 1.858, gfs:29.75,-95.75 (24.957 km) 1.217, gfs:29.50,-95.25 (32.471 km) 1.969, gfs:29.50,-95.75 (32.583 km) 1.585, gfs:30.00,-95.50 (33.831 km) 1.103, gfs:30.00,-95.25 (41.501 km) 1.368, +2 more
openaq	Ozone (ozone)	311 (0.006 km) 0.02051, 312 (8.347 km) 0.01968, 1319333 (13.167 km) 0.02475, 287 (13.542 km) 0.02153, 319 (15.426 km) 0.02107, 325 (19.781 km) 0.005933, 3055 (22.069 km) 0.01884, 3378 (23.726 km) 0.01653, +9 more
openaq	PM10 (pm10)	15852419 (19.314 km) 18.32, 13380082 (23.726 km) 45.89, 271 (35.926 km) 23.48
openaq	PM2.5 (pm25)	8967119 (3.902 km) 8.053, 15588192 (7.397 km) 7.042, 13955816 (7.399 km) 7.272, 15588193 (7.44 km) 7.18, 15399615 (9.038 km) 4.396, 8967093 (10.178 km) 7.121, 15121349 (11.202 km) 2.52, 8966939 (11.337 km) 7.13, +62 more
openweather	Cloud cover (cloud_cover)	grid:west (14.405 km) 54.14, grid:center (14.417 km) 53.26, grid:south (21.745 km) 54.96, grid:north (34.501 km) 47.44, grid:east (38.16 km) 52.18
openweather	Relative humidity (humidity)	grid:west (14.405 km) 72.6, grid:center (14.417 km) 71.45, grid:south (21.745 km) 73.28, grid:north (34.501 km) 69.71, grid:east (38.16 km) 76.43
openweather	Precipitation rate (precipitation)	grid:west (14.405 km) 0.001667, grid:center (14.417 km) 0, grid:south (21.745 km) 0, grid:north (34.501 km) 0, grid:east (38.16 km) 0.003056
openweather	Surface pressure (pressure)	grid:west (14.405 km) 1013, grid:center (14.417 km) 1012, grid:south (21.745 km) 1013, grid:north (34.501 km) 1013, grid:east (38.16 km) 1013
openweather	Air temperature (temperature)	grid:west (14.405 km) 30.85, grid:center (14.417 km) 31.42, grid:south (21.745 km) 30.67, grid:north (34.501 km) 31.15, grid:east (38.16 km) 30.69
openweather	Wind direction (wind_direction)	grid:west (14.405 km) 184.9, grid:center (14.417 km) 198.2, grid:south (21.745 km) 221.1, grid:north (34.501 km) 149.3, grid:east (38.16 km) 188.1
openweather	Wind speed (wind_speed)	grid:west (14.405 km) 1.422, grid:center (14.417 km) 2.244, grid:south (21.745 km) 1.972, grid:north (34.501 km) 2.298, grid:east (38.16 km) 3.544
purpleair	PM2.5 (pm25)	186315 (2.576 km) 8.84, 46237 (5.405 km) 0.2014, 166451 (6.9 km) 8.174, 28851 (9.308 km) 9.163, 293735 (10.612 km) 9.456, 51959 (11.334 km) 7.844, 293725 (11.536 km) 9.366, 301969 (11.717 km) 9.751, +64 more
tceq	Carbon monoxide (co)	482011052 (17.021 km) 0.2953, 482011035 (23.713 km) 0.1429, 482011039 (35.927 km) 0.1115
tceq	Nitrogen dioxide (no2)	482010055 (0.0 km) 2.021, 482011066 (2.946 km) 5.479, 482010047 (15.427 km) 3.389, 482011052 (17.021 km) 7.894, 482010416 (19.781 km) 4.615, 480391004 (22.053 km) 1.533, 482011035 (23.713 km) 6.813, 482010024 (28.285 km) -1.008, +5 more
tceq	Sulfur dioxide (so2)	482010051 (8.334 km) -0.146, 482011035 (23.713 km) -0.173, 482011039 (35.927 km) 0.2607


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (9)
openaq (90)
purpleair (72)
tceq (14)
anomaly
10 km
20 km
30 km
40 km
50 km
Latitude (degrees north)
Longitude (degrees east)
30.1
30.0
29.9
29.8
29.7
29.6
29.5
29.4
−95.8
−95.6
−95.4
−95.2
−95.0


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 3.60 m/s
noaa_gfs
longest arrow = 4.07 m/s
openweather
longest arrow = 3.84 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The anomaly occurred on August 1st, 2026, around 06:01:54 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

The combination of high PM2.5 and NO2 levels is consistent with emissions from industrial activities and vehicle traffic.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

The observed tceq NO2 value of 13.9 ppb exceeded the expected value of 2.2351724137931037 ppb by a z-score of 6.4298105113291175, and the anomaly was classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

Weak surface winds averaging 2 to 3.6 m/s from the south-southwest limited horizontal dispersion of localized NO2 emissions around 06:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

The pollutant with the anomaly is nitrogen dioxide measured by tceq.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

Houston nighttime meteorology at the anomaly time favored near-surface trapping of emissions because humidity was very high, winds were weak to modest, skies were mostly clear, and boundary-layer height was low at roughly 541 m.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

At the time of the NO2 anomaly on 2026-08-01 at 06:00 UTC, nearby ASOS meteorological stations observed south-southwesterly surface winds averaging around 2.2 m/s to 3.6 m/s.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.318744326014901 m/s; n=361 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The tceq NO_2 monitor at 29.696, -95.499 recorded 13.9 ppb at 2026-08-01T06:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

TCEQ air monitoring observations recorded a NO_2 spike of 13.9 ppb at 06:00 UTC while regional SO_2 levels remained near baseline at -0.19 ppb, supporting localized mobile or clean combustion emissions rather than heavy industrial sulfur-rich events.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

A broader regional combustion buildup is only weakly supported because ozone was low near 0.013 ppm at 06:00 UTC, TCEQ carbon monoxide near the anomaly was only about 0.2 ppm and not elevated relative to nearby-period means, particulate matter was low in both openaq and purpleair, and sentinel5p columns showed no coincident large-scale NO_2 or SO_2 enhancement at the available overpass times.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

The PM2.5 levels are elevated at 3.7 ug/m3, which is higher than the mean of 8.0963 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

A local primary NOx emission plume is strongly supported because the anomalous TCEQ NO2 monitor measured 13.9 ppb at 06:00 UTC while the same monitor’s mean in the analysis window was only 2.021 ppb and the 6-hour network mean at 06Z was only 3.228 ppb, indicating a spike concentrated at one site rather than a uniform citywide increase.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.318744326014901 m/s; n=361 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5 and NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

A severe thunderstorm or tornado event in the Houston area has stirred up dust and debris, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

Nighttime stagnation and reduced vertical mixing likely amplified near-surface NO2 concentrations at 06:00 UTC because humidity was very high near 84 to 92 percent, cloud cover was minimal near 1 to 5 percent, and GFS planetary boundary layer height was relatively shallow near 541 m with weak surface winds around 0.45 to 3.6 m/s, conditions that favor accumulation of fresh combustion emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

A sharp reduction in planetary boundary layer height down to approximately 400 meters overnight restricted vertical pollutant mixing, concentrating ground-level nitrogen dioxide.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

ASOS surface observations show wind speeds decreasing to a regional mean of 2.24 m/s with relative humidity rising above 90% at 06:00 UTC on August 1, 2026, facilitating local pollutant stagnation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

Regional photochemical or combustion buildup was not the dominant cause of the spike because ozone was low near 0.013 ppm, PM2.5 was low, CO was not elevated, and the strongest anomaly signal was confined to one TCEQ NO2 monitor rather than broadly reflected across co-pollutants.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

Industrial activities and vehicle traffic are likely contributing factors to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

Local nocturnal combustion sources, such as vehicular traffic or nearby industrial operations upwind of the monitoring site, released nitrogen dioxide into a stable surface layer.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

Regional ambient ozone concentrations dropped to an average of 0.012 ppm around 06:00 UTC on August 1, 2026, reflecting typical nocturnal titration kinetics under limited vertical atmospheric dispersion.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

At 06:00 UTC on August 1, 2026, a TCEQ monitoring station in Houston recorded an elevated nitrogen dioxide (NO_2) level of 13.9 ppb compared to an expected baseline of 2.24 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

The anomaly is due to a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to the accumulation of pollutants in the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

The air quality anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

NOAA GFS model data indicates that the planetary boundary layer height near the anomaly location lowered to approximately 540.5 meters around 2026-08-01T06:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

A severe nitrogen dioxide (NO_2) anomaly was recorded by TCEQ at 2026-08-01T06:00:00+00:00 at coordinates (29.696, -95.499) with an observed concentration of 13.9 ppb compared to an expected baseline of 2.24 ppb (z-score 6.43).

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

A localized primary NOx emission plume from nearby industrial or port-related combustion is a leading hypothesis for the 13.9 ppb NO_2 spike because the anomaly was severe at one TCEQ monitor while the 6-hour network mean remained much lower at 3.228 ppb, indicating spatially confined enhancement rather than a basin-wide event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

Nocturnal accumulation and weak dispersion are supported because humidity was very high near the anomaly at about 83.7 to 92 percent, surface winds were light to modest near 2.24 to 3.60 m/s, skies were nearly clear, and GFS planetary boundary layer height near the anomaly was only about 541 m at 06:00 UTC after a nighttime 6-hour mean of about 413 m, which is consistent with limited vertical mixing that can amplify local NOx emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

NOAA GFS model data indicates that the planetary boundary layer height decreased to a 6-hour mean of approximately 413 meters around 06:00 UTC on August 1, 2026, creating vertical confinement that trapped surface emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Fresh mobile-source combustion from nearby roadway or diesel traffic is a plausible contributor to the NO2 spike because ozone was low near 0.013 ppm at the same time, consistent with fresh NOx-rich emissions that can suppress local ozone through titration, although the absence of a coincident CO spike suggests traffic alone may not fully explain the event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

The TCEQ monitor at 29.696, -95.499 measured 13.9 ppb NO2 at 2026-08-01T06:00:00+00:00, far above its expected 2.235 ppb level, which indicates a severe and strongly localized NO2 enhancement at that site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

Meteorological conditions around 06:00 UTC on August 1, 2026, featured light surface winds of 2.24 m/s and a shallow planetary boundary layer height of approximately 540 m near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.